

Resume

Shubhansh Patel

Systems engineer architecting local-first AI agents and low-level Linux utilities

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About Me

A passionate and driven first-year Computer Science Engineering student with a strong foundation in multiple programming languages and a keen interest in software development, system automation, and innovative project creation. A self-motivated learner with a proven ability to translate complex ideas into functional, cross-platform applications.



Education

Bachelor of Technology, Computer Science &

Engineering LNCT Bhopal | *Expected Graduation: 2028*

Higher Secondary Education (Class XII)

Aadarsh Gyan Jyoti School, Lolri | *Completed: 2024*

- **Score:** 83.6%

High School Education (Nursery - Class X)

Carmel School, Kareli | *Completed: 2022*

- **Class X Score:** 83.6%
- **Academic Honors:** Received **Medal of Merit** (A+ grade in finals) consecutively for Grades 5, 6, 7, and 8.

Experience

Freelance Software Developer

Jan 2024 – Present

- Developed custom automation scripts and desktop utilities for various clients.
- Built and integrated APIs for music management and system bridge tools.
- Provided technical consultation on Linux-based server setups and Android system modifications.

Linux System Specialist & Power User

2017 – Present (7+ Years)

- Extensive experience managing and customizing various Linux distributions.
- Proficient in system administration, kernel compilation, and shell automation across:
 - **RedHat (RHEL) / Fedora**
 - **Arch Linux** (Daily driver for advanced customization)
 - **Debian / Ubuntu / Kali Linux**
 - **Linux From Scratch (LFS)**

Technical Skills

- **AI & Machine Learning:** PyTorch, NumPy, Matplotlib, Artificial Intelligence (LLMs & Agentic Frameworks) — *Learning Phase*
- **Programming Languages:** C++, Java, C, Python, Rust, Kotlin, HTML, CSS, JavaScript (Learning)
- **Operating Systems:** Linux (LFS, Arch, RHEL, Debian, Kali, Fedora), Android (AOSP Modding & Kernel Tweaking)
- **Scripting & Automation:** Bash, Python, C scripting
- **Tools & Concepts:** Git/GitHub, Linux CLI, Android SDK (adb, fastboot), Networking (TCP/IP), Bluetooth HID Protocol, Spotify API, Wine & Proton

Achievements

- **IIT Bombay Techfest (Zonals) (2024):** Designed and built a Gripper Bot and a Meshmerize (Line Follower) bot for competitive robotic trials.
- **Jal Shakti Hackathon (2025):** Participated in a national-level hackathon developing technological solutions for water conservation.
- **Linux From Scratch (LFS):** Compiled a functional Linux operating system from source code, demonstrating mastery of OS architecture.

Projects

Autonomous AI Agent (Ongoing)

- Developing a serverless AI agent from scratch designed for secure, local-first OS management.
- Features full system permissions to manage files and processes, self-correction logic (seeking help when stuck), and the ability to collaborate with other agents.
- **Technologies:** Python, AI Agentic Frameworks, Linux System APIs, Local LLM Integration

Unified Device Bridge (Woolly Wink)

- Developed a personal ecosystem to integrate Android and Debian Linux devices over Wi-Fi and USB.
- Features include two-way clipboard sharing, remote input, notification/call sync, audio

sinking, and remote shell access.

- **Technologies:** Kotlin, Python, C, Android SDK, Linux System APIs, TCP/IP Sockets

Android HID Emulator (In Progress)

- Building an application to emulate a Human Interface Device (HID) using Kotlin and low level C structs for high performance.
- **Technologies:** Kotlin, C, Android NDK, Bluetooth HID Protocol

GitHub Repositories & Open Source

- **Minecraft Modding (Java):** Optimization mods for TNT entities and custom in-game command panels.
- **Antique Record Player:** A stylized Spotify-integrated music player built with Python.
- **System Utilities:** Collection of Rust and C-based CLI tools for Linux workflow optimization.

Hobbies & Interests

- **Academic Reading:** Avid reader of math and science literature: *Calculus* (Apostol), *Mathematical Analysis* (Rudin), *Linear Algebra* (Axler), and *Physics* (Halliday & Walker).
- **Literature & Self-Help:** Works by Fyodor Dostoevsky, Franz Kafka, and Harry Potter; *Atomic Habits* and *Moonwalking with Einstein*.
- **Problem Solving:** Competitive Rubik's Cube solver (PBs: 3x3 in 12s, 2x2 in 3s), Chess player, and Robotics enthusiast.

Declaration

I hereby declare that the above information is true to the best of my knowledge.